

Scholars High School Physics 1: Mechanics (4926)

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Fri

Sep 11, 2026 - Mar 12, 2027

7:30 PM - 9:00 PM ET (4:30 PM - 6:00 PM PT)

Homework

Week: 1 2 **3** 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24

Homework: Week 3



You have completed **0** of **11** challenge problems.

You have **1** writing problem to complete.

Due **Oct 2**.

[Request Writing Problem Extension](#)

Challenge Problems

Problem 1 (43711)



Problem:

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This problem asks you to start applying ideas you learned about projectile motion and reference frames in a qualitative setting.

Suppose you are walking along, holding an empty bottle in your hand. You see a recycling bin at ground level, and want to drop the bottle into the bin. You plan to drop the bottle without throwing it forward or backward at all.

When should you release the bottle?

a) when it's exactly over top the bin

b) when you're approaching the bin, but still a little behind it

c) shortly after you've passed the bin

Problem 2 (46007)

**Problem:**[Report Error](#)

The next two problems on the set review core concepts about projectiles while letting you practice with terminology and notation for motion.

Fill in the blank below with the single best choice.

The variables are:

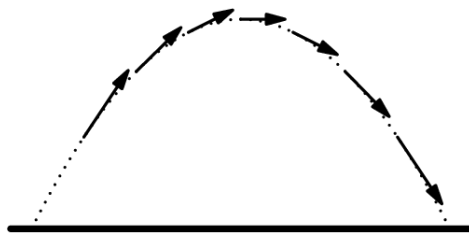
variable	meaning
v	speed
v_x	horizontal component of velocity
v_y	vertical component of velocity
a	magnitude of acceleration
a_y	vertical component of acceleration

When a projectile reaches the highest point in its trajectory, must be zero.

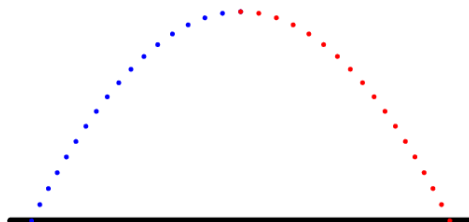
Problem 3 (46008)

**Problem:**[Report Error](#)

A projectile is launched from ground height and travels along a parabola from left to right, landing at the same height it was launched from.



The trajectory can be divided into two halves. Below, the first half is in blue and the second half is in red.



Set the vertical coordinate to be y with $y = 0$ at ground height. y increases as you go up.

Let the time for the projectile to travel the first half be t_1 and the time for it to travel the second half t_2 .

Ignore the first, last, and top points of the trajectory. Use Galileo's model of projectile motion.

y is in the first half and in the second half.

v_y is in the first half and in the second half.

a_y is in the first half and in the second half.

$t_2 - t_1$ is .

 positive

 negative

 zero

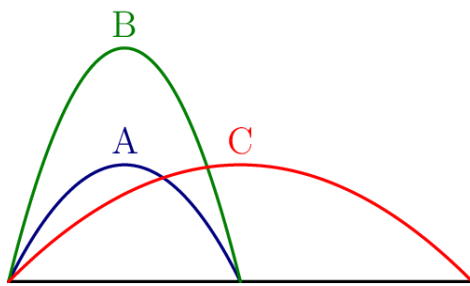
Problem 4 (43736)

**Problem:**[Report Error](#)

Here, you'll practice looking at a drawing and working backwards to figure out what the drawing means in terms of basic physical quantities, but you won't have to do any calculations.

Three projectiles are fired from ground level. Their trajectories are shown below. Air resistance can be ignored. Rank them in order of their horizontal component of velocity, v_x .

If two are equal, drag them next to each other in the middle line (and put the blank on either the top or bottom line as appropriate). If none are equal, leave the blank where it is now.

highest v_x :

lowest v_x :

A

B

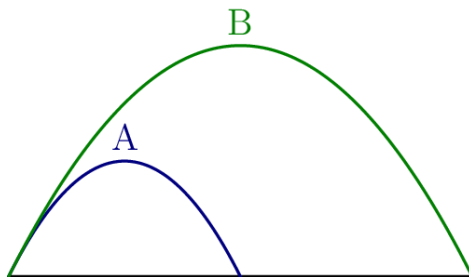
C

Problem 5 (43737)

**Problem:**[Report Error](#)

This is our first quantitative problem on this homework set. It is a good chance to practice proportional reasoning and analyzing horizontal and vertical motions separately, then using information from one analysis in the other.

Two projectiles are fired from ground level. Projectile B goes twice as high and twice as far as projectile A.



The horizontal and vertical velocities of the projectile A, at the moment it's fired, are v_{xA} and v_{yA} . Find the horizontal and vertical velocities of projectile B at the moment it's fired, in terms of those of projectile A.

Give your answers as decimals to the nearest tenth.

$$v_{xB} = \boxed{} v_{xA}$$
$$v_{yB} = \boxed{} v_{yA}$$

Problem 6, Part A (43753)

**Problem:**[Report Error](#)

In this multi-part problem, you'll find the speed of water shot out of a fountain. This first part is a chance to practice visualizing parabolas and thinking about where the parabolic model of motion will start to break down.

The Centennial Fountain shoots water over the Chicago River in Chicago. In the photo below, each tick mark represents one meter.

Use the slider bars to adjust the red parabola. Find the equation that best fits the arc of the water *in the portion that should fit Galileo's model of projectile motion the best*.

You will not be able to fit the entire arc perfectly. Consider whether a parabola will be a better model of the water on the right-hand side or the left-hand side of the fountain, and why. Prioritize fitting the parabola to the side that you expect has the least deviation from Galileo's model.



$$\frac{y}{10 \text{ m}} = \left(-0.03 \frac{x}{10 \text{ m}} \right)^2 + 0.1$$

Problem 6, Part B (43754)

**Problem:**[Report Error](#)

This part of the problem lets you practice algebra skills in physical modeling.

Because the horizontal component of velocity is constant in projectile motion, an equation for the horizontal position over time of some small piece of water shot out of the fountain is

$$x = x_0 + v_{x0}t.$$

And an equation for the vertical position over time is

$$y = y_0 + v_{y0}t - \frac{1}{2}gt^2.$$

The initial positions x_0 and y_0 only depend on our choice of origin, so they don't model any useful information about the speed of the water. We can also choose the time $t = 0$ to be when the water is moving horizontally, so $v_{y0} = 0$. That lets us simplify to

$$\begin{aligned} x &= v_{x0}t \\ y &= y_0 - \frac{1}{2}gt^2 \end{aligned}$$

Find y as a function of x .

Then, assuming $g = 10 \text{ m/s}^2$ and using your answer from part (A), find v_{x0} , the horizontal speed of the water.

You should be accurate to within 1 m/s . Give your answer to the nearest m/s .

$$v_{x0} = \boxed{} \text{ m/s}$$

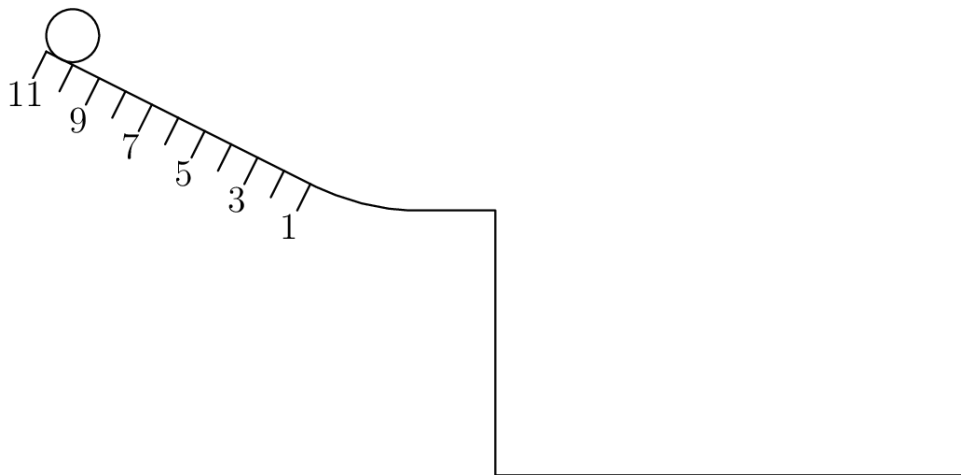
Problem 7 (43740)

**Problem:**[Report Error](#)

This problem is qualitative, and explores an application of projectile motion. The challenge is in figuring out what all the different pieces of the problem mean, not in doing a complicated calculation.

In principle, we can find square roots of numbers by dropping objects. The time it takes an object to fall from height h to the ground obeys $t \propto \sqrt{h}$. To calculate $\sqrt{7}$, for example, we drop two objects, one from 7 times as high as the other, and time both their falls. The ratio of those times would be $\sqrt{7}$.

Accurate timing is difficult, so let's explore another square root calculator.

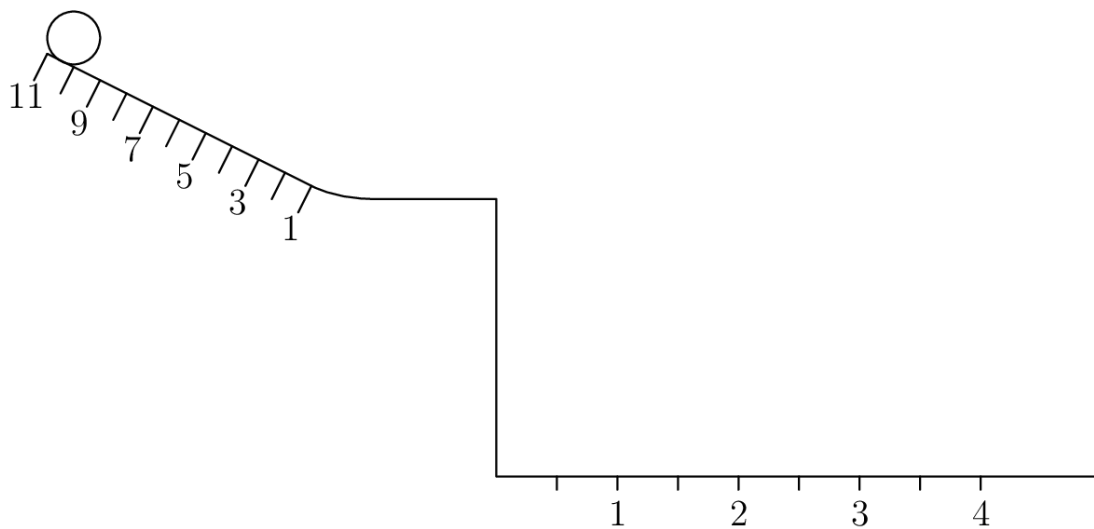


The markings are evenly-spaced. The ball starts at a marking, rolls down, and leaves the table horizontally. It falls in a parabola to the ground.

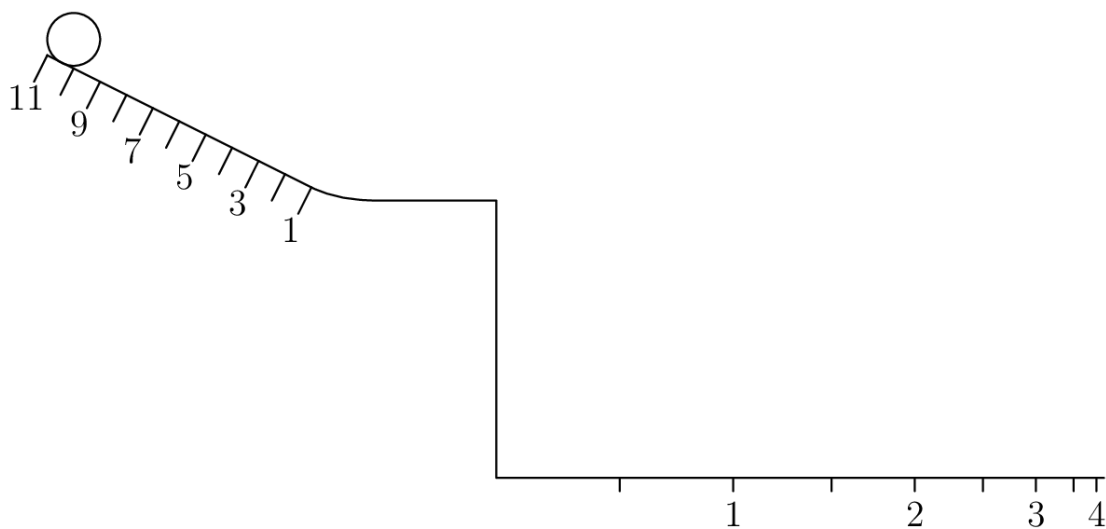
The markings on the ground are supposed to be set up so that the ball hits the square root of the number it was released from. For example, if the ball is released from marking 9 , it should hit marking 3 on the ground.

Which choice shows how the markings should be placed on the floor?

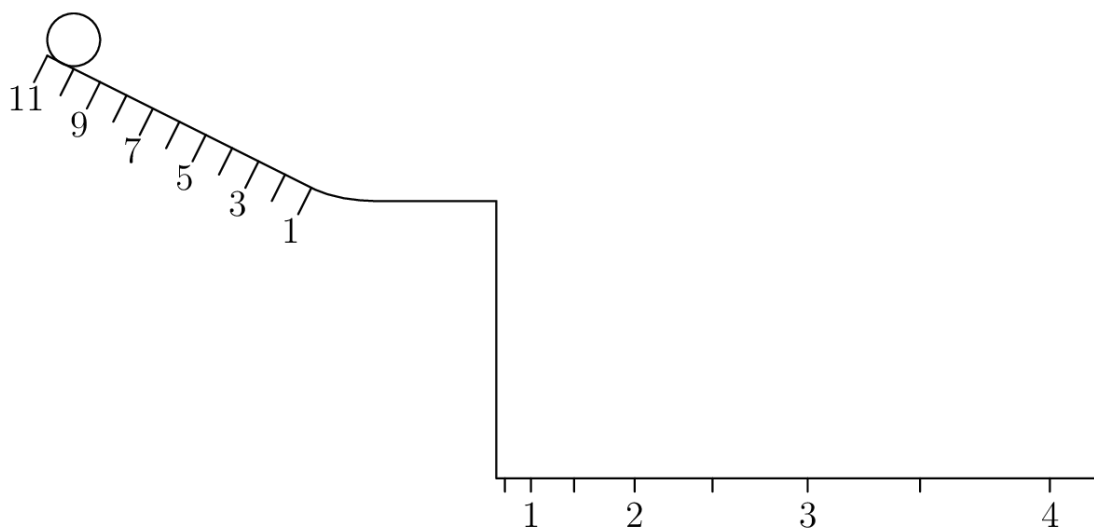
Choose based on the general pattern in the markings, not quantitative measurements of the exact distances.



(A) - Evenly Spaced



(B) - Spacing Decreases to the Right



(C) - Spacing Increases to the Right

A

B

C

none are correct

Problem 8 (43751)

**Problem:**[Report Error](#)

NASA trains astronauts to experience weightlessness with an airplane which flies in a parabolic arc with constant acceleration g toward the ground. Similar planes can be used for civilian purposes, such as this music video:



OK Go, "Upside Down and Inside Out"

The plane can remain on this trajectory for at most 25 s, due to the large change in altitude required. If instead of simulating weightlessness, NASA wanted to fly a trajectory that would simulate the gravitational acceleration of Mars, for what length of time can the plane simulate Mars gravity? Assume that the maximum change in altitude is the same for both trajectories, and that free fall acceleration on Mars is 37% as high as on Earth.

Give your answer to the nearest second.

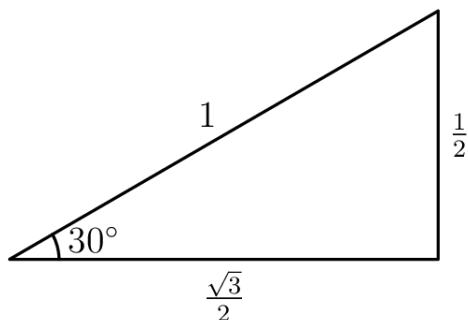
time at Mars free fall acceleration = s

Problem 9 (43757)

**Problem:**[Report Error](#)

This problem lets you use quantitative skills in multiple steps, and think about special triangles.

You may want to remember the $30^\circ - 60^\circ - 90^\circ$ special triangle.



A spring-loaded launcher can shoot a projectile to a height h if launched straight up. If the same launcher points at an angle of 30° above the horizontal and shoots the projectile at the same initial speed, how far will the projectile travel horizontally before landing?

You can ignore air resistance for this problem. Give your answer as a decimal to the nearest hundredth in terms of h .

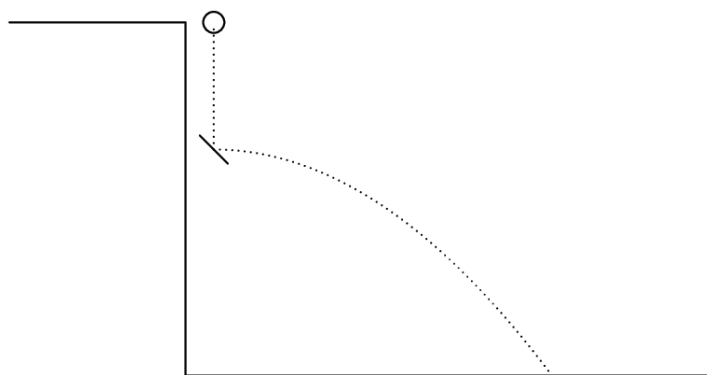
horizontal distance =

Problem 10 (43749)

**Problem:**[Report Error](#)

This challenging quantitative problem asks you to combine what you've learned about acceleration with ideas about horizontal and vertical velocities for trajectories.

You plan to drop a ball off a cliff. The ball will fall straight down for some distance, then bounce off a hard angled plate. When the ball bounces, the plate will redirect it from falling vertically to shooting perfectly horizontally without changing the ball's speed. Then the ball will follow a parabola to the ground.

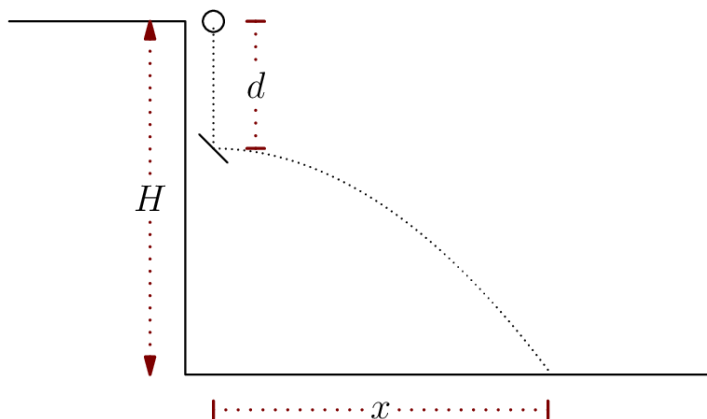


Define the variables:

H : height of cliff

d : distance the ball falls before hitting the plate

x : distance ball travels horizontally



The plate can move up or down, meaning you can choose d .

In terms of H , what value of d should you choose in order to maximize x ?

Give your answer as a decimal to one decimal place.

$$d_{\max} = \boxed{} H$$

Problem 11 (43756)

**Problem:**
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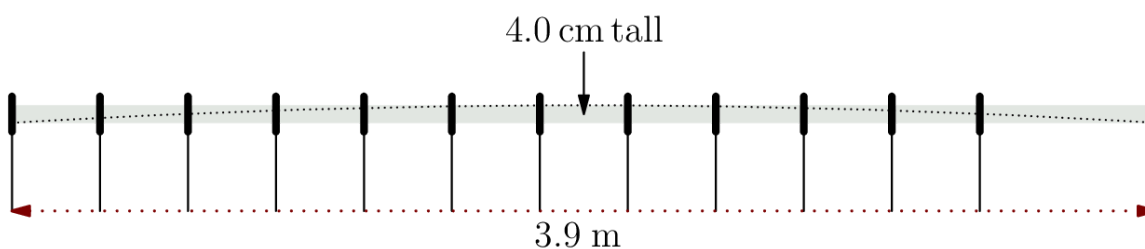
In the ancient Greek poem *The Odyssey*, the hero Odysseus has to shoot an arrow through a series of twelve axes set up in a row. (There are holes in the axes for an arrow to go through.)

In 2019, the TV show *MythBusters Jr.* replicated the feat.



Science Channel, "Replicating the 'Impossible' Odysseus Arrow Shot | Mythbusters Jr.," <https://www.youtube.com/watch?v=9iFuSle2Qkl>

Below is a side view of the axes, plus the path the arrow must follow in a dotted parabola. The arrow must stay in a narrow vertical zone to travel through all the axes.



Model the arrow as a **point**.

To travel through all 12 axes, the arrow needs to stay within a region **4.0 cm** high while traveling a horizontal distance **3.9 m**.

These parameters should let you find a minimum speed of the arrow.

A longbow like Odysseus's might have been able to shoot an arrow **300 m** when shot at an angle of **45°**.

Would Odysseus's longbow shoot the arrow fast enough to get through the axes? Justify your answer quantitatively. You may ignore air resistance on the arrow.

Your Response:

Attachments: You may include images and files (up to 1 MB) with this submission. Even if you upload files, please enter a text explanation.

Writing Problem:

This is a **free response** question. Your response will be graded by a real living person! Please explain your answer fully, writing in complete sentences. For tips on how to write a good solution, see [here](#). Your submission will be finalized the evening of Oct 2.

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Typesetting math: 100%

